

University of the South Pacific

School of Information Technology, Engineering, Mathematics, and Physics

IS328 - Data Mining

Short Test Solutions

Semester 2, 2024

Possible Marks:	30
Weight:	5%
Time Allowed:	50 minutes
Number of Pages:	7 (including this page)

Learning Outcomes:

1. Explain the concepts, definitions and processes of Data Mining

Instructions:

1. Answer ALL questions.
2. Write ALL the answers in the spaces provided in the Answer Booklet.
3. If you require extra space for your answers, use the last page.

Q1: Pre-Processing (20 marks)

- a) What is the difference between Data Mining and Knowledge Discovery in Databases (KDD)? [2 marks]

Data Mining refers to the specific step in the KDD process that **involves extracting patterns**, trends, and useful information from large datasets using statistical, machine learning, and database management techniques. Knowledge Discovery in Databases (KDD), on the other hand, is the overall process of **discovering useful knowledge from data**, which includes data preparation, data selection, data cleaning, and data mining, followed by the evaluation and interpretation of the results.

- b) Explain the role of feature selection in Data Mining and why it is important. [3 marks]

Feature selection is the process of selecting a subset of relevant features (variables, predictors) for use in model construction. It is important because it helps reduce the dimensionality of the data, which can improve model performance, reduce computational cost, and enhance model interpretability by focusing on the most significant variables.

- c) What is the difference between supervised and unsupervised learning in Data Mining? Provide an example of each. [4 marks]

Supervised learning involves training a model on a labeled dataset, where the outcome variable (target) is known. The model learns to predict the target based on input features. An example **is classification**, where the model predicts categories, like spam or not spam in an email dataset. Unsupervised learning, in contrast, involves training a model on a dataset without labeled outcomes. The model tries to find patterns or structures in the data. An example is **clustering**, where the model groups data points into clusters based on their similarities.

- d) Explain the concept of binning in data preprocessing and its purpose. [2 marks]

Binning is a data preprocessing technique where continuous data is divided into a number of intervals, or "bins." The purpose of binning is to reduce the effect of minor observation errors and to smooth out noise in the data, making patterns easier to detect.

- e) Given the following data [3, 7, 8, 15, 24, 28, 33, 42, 48], perform binning using equal-width binning with three bins. Show the bin boundaries and the data points in each bin. [4 marks]

Step 1: Determine the range:

$$\text{Range} = 48 - 3 = 45$$

Step 2: Determine bin width:

$$\text{Bin Width} = 45 / 3 = 15$$

Step 3: Define bins:

- Bin 1: [3, 18)
- Bin 2: [18, 33)
- Bin 3: [33, 48]

Step 4: Place data into bins:

- Bin 1: 3, 7, 8, 15
- Bin 2: 24, 28
- Bin 3: 33, 42, 48

- f) Given the dataset [4, 8, 15, 16, 23, 42], normalize it using Min-Max normalization.

[5 marks]

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

First, find the minimum and maximum values:

$$\min = 4, \max = 42$$

Then, apply the Min-Max normalization formula to each value:

- For 4: $x' = \frac{4-4}{42-4} = 0$
- For 8: $x' = \frac{8-4}{42-4} = \frac{4}{38} \approx 0.105$
- For 15: $x' = \frac{15-4}{42-4} = \frac{11}{38} \approx 0.289$
- For 16: $x' = \frac{16-4}{42-4} = \frac{12}{38} \approx 0.316$
- For 23: $x' = \frac{23-4}{42-4} = \frac{19}{38} = 0.5$
- For 42: $x' = \frac{42-4}{42-4} = 1$

Q2: Classification Techniques (10 marks)

- a) Describe how a decision tree is constructed. List one criteria used to split the nodes?

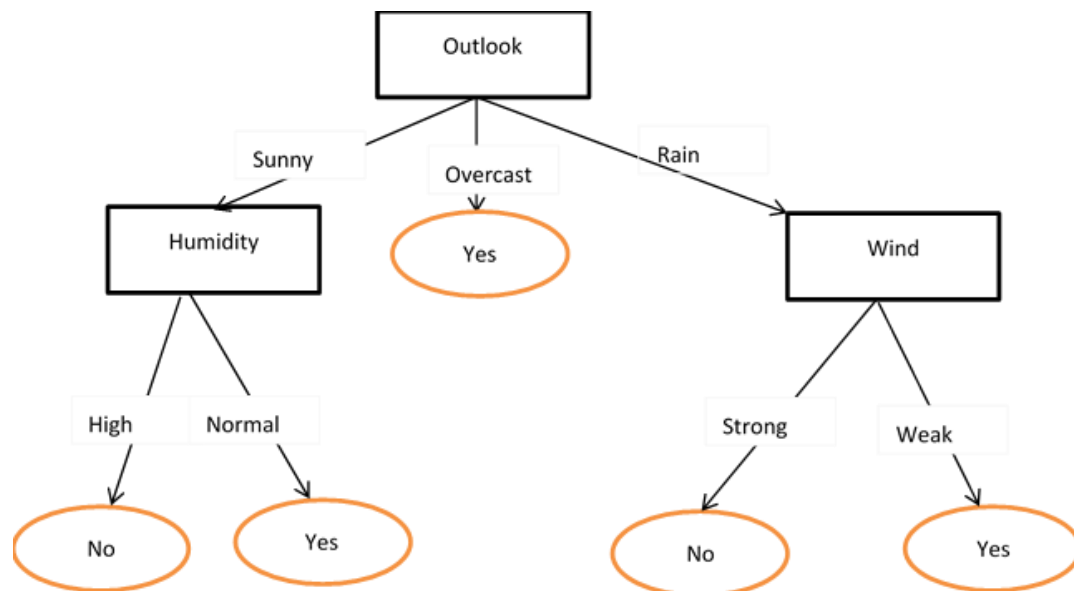
[3 marks]

A decision tree is constructed by recursively splitting the dataset into subsets based on the attribute that provides the most significant information gain or the greatest reduction in entropy.

The criteria for splitting nodes include:

- Information Gain: Measures how much a feature contributes to reducing uncertainty about the output.
- Gini Index: Measures the impurity of a node. Lower Gini Index indicates better splits.
- Chi-Square Statistic: Tests the independence of two categorical variables, used for categorical data.

- b) Based on the decision tree below, generate all possible IF-THEN rules that can be used to predict the outcomes. [2.5 marks]



Logic Rules from the Decision Tree:

1. Rule 1:
IF Outlook = Sunny AND Humidity = High
THEN Play Tennis = No
2. Rule 2:
IF Outlook = Sunny AND Humidity = Normal
THEN Play Tennis = Yes
3. Rule 3:
IF Outlook = Overcast
THEN Play Tennis = Yes
4. Rule 4:
IF Outlook = Rain AND Wind = Strong
THEN Play Tennis = No
5. Rule 5:
IF Outlook = Rain AND Wind = Weak
THEN Play Tennis = Yes

c) State one advantage of using decision trees for classification. *[0.5 mark]*

1. Easy to understand.
2. Map nicely to a set of production rules.
3. Applied to real problems.
4. Make no prior assumptions about the data.
5. Able to process both numerical and categorical data.

d) Suppose two models (model 1 and model 2) are being tested for granting entry to a secure sensitive area using face verification system. The results are presented via confusion matrices as below. Calculate the accuracy and error rate of the two models and explain which model should be adopted, providing justification for your choice. *[4 marks]*

Model 1	Predicted	
	Yes	No
Actual		
Yes	1500	50
No	200	500

Model 2	Predicted	
	Yes	No
Actual		
Yes	1500	200
No	50	500

Model 1:

- True Positive (TP): 1500
- False Positive (FP): 200
- False Negative (FN): 50
- True Negative (TN): 500

Model 2:

- True Positive (TP): 1500
- False Positive (FP): 50
- False Negative (FN): 200
- True Negative (TN): 500

Accuracy Calculation

Accuracy is calculated using the formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total number of predictions}}$$

Model 1 Accuracy:

$$\text{Accuracy} = \frac{1500 + 500}{1500 + 200 + 50 + 500} = \frac{2000}{2250} \approx 0.8889 \text{ or } 88.89\%$$

Model 2 Accuracy:

$$\text{Accuracy} = \frac{1500 + 500}{1500 + 50 + 200 + 500} = \frac{2000}{2250} \approx 0.8889 \text{ or } 88.89\%$$

Error Rate Calculation

Error rate is calculated using the formula:

$$\text{Error Rate} = 1 - \text{Accuracy}$$

Model 1 Error Rate:

$$\text{Error Rate} = 1 - 0.8889 = 0.1111 \text{ or } 11.11\%$$

Model 2 Error Rate:

$$\text{Error Rate} = 1 - 0.8889 = 0.1111 \text{ or } 11.11\%$$

Conclusion

Both models have the same accuracy and error rate. However, the choice between the models should be based on the consequences of **false positives** and **false negatives**:

- **Model 1** has more false positives (200) but fewer false negatives (50). This makes it more lenient in granting access, which is a concern in a high-security environment.
- **Model 2** has fewer false positives (50) but more false negatives (200). This makes it stricter, leading to fewer unauthorized entries, which is preferred in a secure setting.

Given the context of securing a sensitive area, **Model 2** is preferred due to its lower false positive rate, thereby reducing the risk of unauthorized access.